

Regime-Conditional Attention Head Selection for Time Series Transformers

Week 7 Interim Status Report

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Project Title:

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1. Project Summary

This project investigates the hypothesis that the importance of attention heads in Transformer-based models used for time series forecasting is not always constant, but may vary across different temporal regimes. Existing attention head pruning methods typically compute a single global importance score for each head and base pruning decisions on this score. However, in time series data, some periods may be trend-dominant, others seasonal, and others residual/noise-dominant. Consequently, each attention head may contribute differently depending on the regime.

As stated in the project proposal, the primary objective of this study is to decompose validation windows into trend, seasonal, and residual components using STL decomposition, to label each window according to its dominant component, and to measure attention head importance separately for each regime. Two different approaches are then evaluated: a static/global pruning policy and a dynamic head-selection policy that varies according to the regime of the input window.

This interim report summarizes the data preparation, model selection, system setup, demo experiments, and initial pruning results completed as of Week 7. It also presents the current versions of the Abstract, Related Work, and Proposed Implementation sections.

2. Abstract

Transformer-based architectures have shown strong performance in long-term time series forecasting; however, their multi-head attention mechanisms may contain redundant heads that increase computational cost. Existing attention head pruning methods generally assign a single global importance score to each head, assuming that head importance is stable across the entire validation set. In this project, we investigate whether attention head importance changes depending on the temporal regime of the input window. We propose a regime-aware attention head selection approach for time series Transformers. First, time series windows are decomposed using STL into trend, seasonal, and residual components. Each window is assigned a dominant regime label based on the component with the highest normalized variance. Then, attention head importance is measured separately for each regime by masking one head at a time and observing the resulting change in validation loss. We evaluate both static pruning and dynamic regime-aware head selection on PatchTST using the ETTh1 dataset with a prediction length of 96. Preliminary results show that attention head importance differs across regimes and that pruning can improve validation performance. However, test-set results indicate that the current masking-based pruning strategy does not yet generalize reliably, motivating future work on random and magnitude-based baselines, fine-tuning after pruning, and efficiency measurements. At the current stage, static pruning and dynamic regime-aware selection improved validation performance, but neither method improved the test performance over the unpruned B4 baseline.

3. Data Collection Status

The dataset acquisition and preparation phase has been completed. The ETTh1 dataset [3] was used for the initial experiments in this project. This choice is consistent with the project proposal, which specifies PatchTST as the main model, ETTh1 as the initial dataset, and a prediction length of 96.

Dataset configuration used:

- Dataset: ETTh1
- Prediction length: 96
- Input sequence length: 336

- Target variable: OT
- Model: PatchTST

The ETTh1 dataset has been prepared in a format compatible with the Time-Series-Library and made usable within the Colab environment. The dataset has been split into training, validation, and test sets. The validation and test windows have each been labeled separately using STL decomposition [5].

3.1 Validation Regime Distribution

After STL decomposition, the windows in the validation set were labeled as follows:

Regime	Window Count	Percentage
Trend	2134	76.62%
Residual	359	12.89%
Seasonal	292	10.48%

This distribution shows that trend-dominant windows are the majority in the validation set. Nevertheless, a sufficient number of samples exist in the seasonal and residual classes to allow for head importance analysis.

3.2 Test Regime Distribution

The same STL-based labeling method was applied to the test set:

Regime	Window Count	Percentage
Trend	2606	93.57%
Seasonal	125	4.49%
Residual	54	1.94%

The distribution in the test set is considerably more imbalanced than in the validation set. The vast majority of test windows are labeled as trend-dominant. This has been taken into account as an important factor when interpreting the test performance of the dynamic regime-aware method.

4. Platform and System Status

Experiments were conducted primarily on Google Colab. A GitHub repository was used for local development and version control. Experiments were executed on Google Colab T4 GPU runtime when available.

4.1 Platforms Used

- Development environment: Google Colab
- Version control: GitHub
- Main repository branch: didem/patchtst-baseline
- Main model implementation: Time-Series-Library
- Dataset source: ETDataset / ETTh1
- Main framework: PyTorch

4.2 Libraries Used

The main libraries used in the project are as follows:

- torch
- numpy
- pandas
- statsmodels
- tqdm
- Time-Series-Library
- sktime
- patool
- reformer-pytorch
- local-attention
- hyper_connections
- axial_positional_embedding
- product_key_memory
- colt5_attention

Since the Time-Series-Library requires several additional attention-related packages when importing certain models, the necessary dependency installations were added to the Colab notebooks, which reduced model import errors.

5. Setup Problems and Solutions

Several setup and environment-related issues were encountered during the project.

5.1 Full Requirements Installation Problem

Installing the entire requirements.txt file of the Time-Series-Library directly caused several dependency conflicts within the Colab environment. Therefore, instead of installing all requirements, only the necessary minimal packages were installed.

The following approach was used as a solution:

```
pip install patool sktime scikit-base --no-deps
```

In addition, the following packages were installed separately for certain model imports:

- reformer-pytorch
- local-attention
- hyper_connections
- axial_positional_embedding
- product_key_memory
- colt5_attention

5.2 Dataset Path Problems

Since the ETTh1 dataset was not always readily available within the Colab runtime, an automatic copy/download step was added to the notebooks. If the dataset is present on Google Drive, it is copied from there; otherwise, it is downloaded from GitHub.

5.3 GitHub and File Tracking

Notebooks, experiment outputs, and CSV results have been added to GitHub. The work has so far been organized into separate notebooks:

- 01_patchtst_baseline.ipynb
- 02_regime_detection.ipynb
- 03_head_importance.ipynb
- 04_static_pruning_25_validation_test.ipynb
- 05_dynamic_regime_aware_selection.ipynb
- 06_dynamic_regime_aware_75_keep.ipynb
- 07_test_regime_detection_dynamic_75_eval.ipynb

On the Git side, branch synchronization and rebasing before pushing were occasionally required. These issues were resolved using `git pull --rebase` followed by `git push`.

6. Demo Runs and Current Experimental Results

At this stage, the main experimental pipeline has been established, and the first round of pruning and dynamic selection experiments has been completed.

6.1 Baseline Model Selection

Several model configurations were tested on PatchTST, varying sequence length, hidden dimension, and the number of attention heads. Based on these comparisons, the B4 model was selected as the main baseline.

B4 model configuration:

- Model: PatchTST
- Dataset: ETTh1
- Prediction length: 96
- Input sequence length: 336
- d_model: 128
- d_ff: 256
- n_heads: 8
- e_layers: 3
- Total heads: 24

Experiment	Seq Len	d_model	Heads	Val Loss	Test MSE	Test MAE
B0	336	128	4	0.6732	0.3739	0.3989
B1	512	128	16	0.7382	0.3764	0.4053
B2	512	256	16	0.7811	0.3947	0.4234
B3	336	128	16	0.6781	0.3735	0.3994
B4	336	128	8	0.6735	0.3725	0.3982

B4 baseline result:

Model	Validation Loss	Test MSE	Test MAE
B4 PatchTST	0.6735	0.3725	0.3982

The B4 model was selected as the main model because it delivers solid test performance while containing a sufficient number of attention heads for pruning experiments.

6.2 STL-Based Regime Detection

Validation and test windows were decomposed into trend, seasonal, and residual components using STL decomposition. For each window, component variances were normalized, and the component with the highest score was assigned as the window's dominant regime label.

This step is directly consistent with the first main implementation step outlined in the project proposal.

6.3 Attention Head Importance Analysis

Each attention head was masked individually, and its effect on validation loss was measured. This procedure was carried out both for the overall validation set and separately for the trend, seasonal, and residual regimes.

An important observation resulted from this analysis: attention head importance varies according to regime.

Some heads appeared useful in trend-dominant windows, while the same head could appear less useful, or even harmful, in seasonal or residual windows. This observation is one of the first findings supporting the project's main hypothesis.

6.4 Static 25% Head Pruning

In the static pruning experiment, the 6 heads with the lowest overall importance score were disabled, corresponding to pruning 25% of the total 24 heads.

Setting	Active Heads	Pruned Heads	Validation MSE	Validation MAE
B4 no pruning	24	0	0.6781	0.5551
Static 25% pruning	18	6	0.6537	0.5507

Static pruning reduced the validation MSE by approximately 3.60%. However, the same improvement was not observed on the test set:

Setting	Test MSE	Test MAE
B4 no pruning	0.3725	0.3982
Static 25% pruning	0.3783	0.4044

Static pruning increased the test MSE by approximately 1.55%. This indicates that the pruning mask selected on the validation set does not generalize perfectly to the test set.

6.5 Dynamic Regime-Aware 50% Keep

In the first dynamic experiment, the 12 most important heads were selected for each regime, and each input window was processed using a different mask according to its own regime label.

Setting	Active Heads	Pruned Heads	Validation MSE	Validation MAE
B4 no pruning	24	0	0.6781	0.5551
Dynamic 50% keep	12	12	0.6678	0.5553

Dynamic 50% keep reduced the validation MSE by approximately 1.51% relative to the baseline. However, since this setting is quite aggressive, degradations were observed in the seasonal and residual windows.

6.6 Dynamic Regime-Aware 75% Keep

To enable a fairer comparison with static pruning, a 75% keep ratio was tested for the dynamic method. In this setting, 18 heads were kept active and 6 heads were disabled for each window, matching the pruning ratio used in static pruning.

Setting	Active Heads	Pruned Heads	Validation MSE	Validation MAE
B4 no pruning	24	0	0.6781	0.5551
Static 25% pruning	18	6	0.6537	0.5507
Dynamic 75% keep	18	6	0.6597	0.5512

Dynamic 75% keep reduced the validation MSE by approximately 2.71% relative to the baseline. However, static pruning still produced better results on the validation set.

On the test set:

Setting	Active Heads	Pruned Heads	Test MSE	Test MAE
B4 no pruning	24	0	0.3725	0.3982
Static 25% pruning	18	6	0.3783	0.4044
Dynamic 75% keep	18	6	0.3794	0.4053

On the test set, the baseline achieved the best performance. Although Dynamic 75% keep provided improvement on residual-dominant windows, the overall test performance did not improve, since the majority of the test set is trend-dominant.

Regime	Baseline MSE	Dynamic 75 MSE	Change
Residual	0.3621	0.3570	-1.42%
Seasonal	0.3411	0.3556	+4.27%
Trend	0.3743	0.3810	+1.79%

7. Related Work

Transformer-based models are widely used in time series forecasting due to their capacity to model long-range dependencies [6]. PatchTST is a strong model that improves the effectiveness of the Transformer architecture for time series forecasting by dividing the series into patches [1]. PatchTST was chosen for this project because its attention mechanism operates over temporal patches, which is naturally compatible with the notion of temporal regimes. The project proposal also specifies PatchTST as the main model. Other recent architectures, such as iTransformer, instead invert the attention mechanism to

operate over variates rather than time steps, illustrating an alternative direction for Transformer-based forecasting design [2].

In the attention head pruning literature, a single importance score is typically computed for each attention head. This global score is measured over the entire validation set, and heads with low scores are pruned, as shown for machine translation and language understanding models by Michel et al. and Voita et al. [7, 8]. However, this approach relies on the assumption that each head is equally important for all input types. This assumption may be limiting for time series data, since different temporal structures such as trend, seasonality, and residual/noise may require different attention behaviors.

Unlike existing global pruning approaches, this project examines head importance values on a per-temporal-regime basis. In this respect, the study combines ideas from attention pruning and regime-aware time series analysis. While the literature contains ideas related to conditional computation and input-dependent model selection, such as dynamic token sparsification in vision transformers [7], this project applies this idea at the level of attention head selection specifically for time series Transformer models.

8. Proposed Implementation

The proposed implementation of the project consists of several stages.

8.1 Baseline Model Training and Selection

In the first stage, the PatchTST model is trained on the ETTh1 dataset with different configurations. The goal is both to obtain a strong forecasting baseline and to select a model suitable for attention head pruning. This stage has been completed, and the B4 model has been determined as the main baseline.

8.2 STL-Based Regime Labeling

Each input window is decomposed into trend, seasonal, and residual components using STL decomposition. The variance of each component is computed and normalized. The component with the highest normalized variance is assigned as the window's regime label.

Formally:

```
regime(window) = argmax { trend_score, seasonal_score, residual_score }
```

This procedure has been applied to both the validation and test sets.

8.3 Regime-Specific Head Importance

In the selected PatchTST model, each attention head is masked individually. After each masking operation, the validation loss is recomputed. The change in loss is used as the head importance score:

```
importance(head) = loss(masked_head) - loss(baseline)
```

This score is computed separately for the overall validation set and for each regime:

- trend_importance
- seasonal_importance
- residual_importance
- overall_importance

This stage has been completed.

8.4 Policy A: Static Pruning

In the static pruning policy, the overall importance value is used for each head. The 6 heads with the lowest importance values are disabled, corresponding to 25% of the total number of heads.

Currently, static pruning is implemented as masking-based inference; that is, heads are not physically removed from the model but are masked at the level of the attention output. In this stage, “pruning” refers to masking the selected attention heads during inference rather than physically removing their parameters.

In a future stage, fine-tuning of the static pruned model is planned. This will complete the Policy A definition outlined in the project proposal, which specifies that the model will be fine-tuned after low-scoring heads are pruned.

8.5 Policy B: Dynamic Regime-Aware Head Selection

In the dynamic policy, the regime label of each input window is first determined. Then, the most important head subset previously computed for that regime is kept active. For example:

- trend-dominant window → trend-specific head mask
- seasonal-dominant window → seasonal-specific head mask
- residual-dominant window → residual-specific head mask

This method has been tested with 50% keep and 75% keep settings. The 75% keep setting uses the same pruning ratio as static pruning, enabling a fairer comparison.

9. Current Project Stage

As of Week 7, the project has completed the majority of the proposed main workflow. The current status is as follows:

Stage	Status
Data collection	Completed
ETTh1 data preparation	Completed
PatchTST baseline training	Completed
Baseline model selection	Completed
STL-based validation regime detection	Completed
STL-based test regime detection	Completed
Regime-specific head importance	Completed
Static pruning initial experiment	Completed
Dynamic regime-aware selection initial experiments	Completed
Initial test evaluation	Completed
Random pruning baseline	In progress
Magnitude-based pruning baseline	In progress
Fine-tuning after pruning	In progress
Latency/FLOPs/efficiency measurement	In progress

Stage	Status
ETM1/Weather extension	Optional next stage

10. Remaining Work and Next Steps

Several additional experiments will be carried out to complete all comparisons specified in the proposal. The proposal states that the method will be compared against no-pruning, random head pruning, global importance pruning, and magnitude-based pruning baselines. So far, no-pruning and global/static importance pruning have been completed. Random pruning and magnitude-based pruning have not yet been completed. The next steps are as follows.

10.1 Random Head Pruning Baseline

A baseline will be constructed by randomly disabling 6 out of 24 heads, using the same pruning ratio. Multiple random seeds will be used for more reliable results.

Plan:

- 5 different random seeds
- 6 random heads disabled per seed
- Computation of validation and test MSE/MAE
- Reporting of mean and standard deviation

10.2 Magnitude-Based Pruning Baseline

The projection weight norm will be computed for each attention head, and heads with low norm values will be pruned. This will complete the magnitude-based pruning baseline specified in the proposal.

10.3 Fine-Tuning After Static Pruning

After static pruning, the model will be fine-tuned for several epochs. This stage is particularly important because the pruning experiments conducted so far are based on masking-based inference. Fine-tuning may allow the pruned model to adapt to the disabled heads.

10.4 Efficiency Metrics

For evaluation, not only forecasting accuracy but also efficiency metrics will be measured. The proposal specifies that, in addition to MSE/MAE, parameter count, FLOPs, and inference latency metrics will be used.

Planned efficiency metrics:

- Active head ratio
- Inference latency
- Theoretical pruned head ratio
- FLOPs estimate
- Parameter count discussion

Note: In the current masking-based pruning implementation, the parameter count is not physically reduced. Therefore, structural pruning or theoretical effective parameter reduction will be separately addressed for changes in parameter count.

10.5 Additional Dataset or Prediction Length

If time permits, the experiments will be extended to the ETTm1 [3] or Weather [4] datasets. The proposal also specifies the possibility of extending to the ETTm1 and Weather datasets after ETTh1.

11. Overall Assessment

Up to this stage, the main pipeline planned in the project proposal has been successfully established. Data preparation, baseline selection, regime detection, head importance measurement, static pruning, and dynamic regime-aware selection experiments have all been completed. The initial results show that attention head importance values change according to temporal regime. This finding supports the project's main hypothesis.

On the validation set, static pruning and dynamic regime-aware selection provided improvement over the baseline. However, on the test set, the unpruned B4 model achieved the best result. This indicates that the current head selection strategy is promising on the validation set but requires further development in terms of generalization to the test set.

In the next stage, the random pruning and magnitude-based pruning baselines will be completed, fine-tuning after static pruning will be performed, and efficiency metrics such as latency and FLOPs will be measured. Once these studies are completed, the project can be evaluated more comprehensively in terms of both forecasting accuracy and model efficiency.

References

- [1] Yuqi Nie et al., A Time Series is Worth 64 Words: Long-Term Forecasting with Transformers, ICLR 2023.
- [2] Yong Liu et al., iTransformer: Inverted Transformers Are Effective for Time Series Forecasting, 2024.
- [3] ETT Dataset, <https://github.com/zhouhaoyi/ETDataset>
- [4] Weather Dataset, https://nixtlaverse.nixtla.io/datasetsforecast/long_horizon2.html
- [5] Cleveland et al., STL: A Seasonal-Trend Decomposition Procedure Based on Loess, 1990.
- [6] Vaswani et al., Attention Is All You Need, 2017.
- [7] Yongming Rao et al., DynamicViT: Efficient Vision Transformers with Dynamic Token Sparsification, NeurIPS 2021.